Electronic Supplementary Material

Early physiological phenotypes and outcomes in critically ill adults with non-traumatic subarachnoid hemorrhage

This Electronic Supplementary Material (ESM) accompanies the Intensive Care Medicine-style main manuscript. It provides cohort algorithms, code lists, variable mappings, extended tables, sensitivity analyses, supplementary figures, and reproducibility details.

ESM 1. Cohort Algorithm and Provenance

ESM Table 1. MIMIC-IV cohort-flow counts

Step	Definition	Admissions	Patients
01	Source SAH admissions	1,576	1,531
02	Adult source SAH	1,576	1,531
03	Adult non-traumatic SAH; no aneurysm evidence required	1,574	1,529
04	First ICU stay among NSAH hospitalizations	1,317	1,298
05	ICU length of stay ≥24 h	1,212	1,197
06	Eight core features with ≤2 missing values	1,186	1,173
07	Primary analysis after excluding massive early transfusion	1,186	1,173
08	Sensitivity cohort: ICU length of stay ≥48 h	1,005	996
09	Sensitivity cohort: no recorded RBC transfusion in 0–48 h	1,162	1,149

ESM Table 2. eICU cohort-flow counts

Step	Definition	Unit stays	Patients
01	SAH candidate unit stays from diagnosis/admissionDx	2,880	2,571
02	Adult patients	2,863	2,554
03	Exclude traumatic SAH flags	1,314	1,158
04	First ICU stay per unique patient	1,153	1,153
05	ICU length of stay ≥24 h	903	903
06	Eight core features with ≤2 missing values	843	843
07	Sensitivity cohort: ICU length of stay ≥48 h	626	626
08	Sensitivity cohort: no recorded RBC transfusion in 0–48 h	819	819
09	Strict SAH sensitivity definition	605	605
10	Low-missing sensitivity: ≤1 missing core feature	819	819
11	Complete-case sensitivity	428	428

Counts in ESM Tables 1 and 2 are derived from `dist/20260708/figures/fig1_cohort_flowchart_data.json`, which was generated from BigQuery intermediate cohort-flow tables.

ESM Table 3. Diagnostic code and text algorithm

Data source	Inclusion evidence	Exclusion evidence
MIMIC-IV	ICD-9 `430` (subarachnoid hemorrhage); ICD-10 `I60.x` (nontraumatic subarachnoid hemorrhage subtypes)	ICD-10 `S06.6x` or diagnostic text consistent with traumatic subarachnoid hemorrhage
eICU-CRD	ICD-9 `430`; `apacheadmissiondx` containing "subarachnoid hemorrhage"; diagnosis table text consistent with subarachnoid hemorrhage	Diagnosis/admission text consistent with traumatic subarachnoid hemorrhage

The algorithm intentionally did not require aneurysm diagnosis or aneurysm-securing procedure evidence for the primary NSAH cohort. A strict aneurysm subgroup was evaluated as sensitivity analysis.

ESM 2. Variable Mapping and Preprocessing

ESM Table 4. Core physiological feature definitions

Domain	Feature	Definition	Window	Primary preprocessing
Neurological	`gcs_motor_min_48h`	Minimum GCS motor score	0–48 h after ICU admission	Z-score standardization
Circulatory	`map_min_48h`	Minimum mean arterial pressure	0–48 h after ICU admission	Z-score standardization
Circulatory	`shock_index_max_48h`	Maximum heart rate / systolic blood pressure	0–48 h after ICU admission	Z-score standardization
Oxygenation	`spo2_min_48h`	Minimum pulse oximetry oxygen saturation	0–48 h after ICU admission	Z-score standardization
Renal	`creatinine_max_48h`	Maximum serum creatinine	0–48 h after ICU admission	log1p transformation, then Z-score
Coagulation	`inr_max_48h`	Maximum international normalized ratio	0–48 h after ICU admission	log1p transformation, then Z-score
Hematologic	`hb_min_48h_all`	Minimum hemoglobin	0–48 h after ICU admission	Z-score standardization
Hematologic	`platelet_min_48h`	Minimum platelet count	0–48 h after ICU admission	Z-score standardization

Missing values were imputed using MIMIC-IV derivation-cohort medians. External validation used frozen MIMIC-IV medians, scaler parameters, PCA loadings, and K-means centroids.

ESM Table 5. Feature availability and data quality

Feature	MIMIC missing N	MIMIC missing %	eICU missing N	eICU missing %
`inr_max_48h`	65	5.48%	446	49.4%
`creatinine_max_48h`	1	0.08%	31	3.4%

Feature	MIMIC missing N	MIMIC missing %	eICU missing N	eICU missing %
`platelet_min_48h`	0	0.00%	75	8.3%
`hb_min_48h_all`	0	0.00%	64	7.1%
`shock_index_max_48h`	0	0.00%	10	1.1%
`map_min_48h`	0	0.00%	10	1.1%
`spo2_min_48h`	0	0.00%	10	1.1%
`gcs_motor_min_48h`	0	0.00%	5	0.6%

ESM Table 6. Principal component loadings for the standardized feature space

Physiological feature	PC1 loading	PC2 loading	PC3 loading
Hemoglobin min	0.385	-0.104	-0.211
GCS motor min	0.412	0.289	0.150
MAP min	0.298	0.415	-0.380
Shock index max	-0.420	-0.210	0.312
SpO2 min	0.198	-0.580	0.412
Creatinine max	-0.311	0.395	0.520
INR max	-0.354	0.224	0.490
Platelets min	0.390	-0.380	-0.154

ESM 3. Extended Tables

ESM Table 7. Baseline characteristics of the MIMIC-IV derivation cohort

Characteristic	Overall (N=1,186)	P1 (n=694)	P2 (n=384)	P3 (n=108)	p-value
Demographics					
Age, median [IQR], years	61.0 [50.0–71.0]	59.5 [49.0–70.0]	62.0 [52.0–75.0]	62.5 [48.0–70.0]	0.026
Female sex, n (%)	665 (56.1%)	393 (56.6%)	234 (60.9%)	38 (35.2%)	<0.001
White race, n (%)	696 (58.7%)	449 (64.7%)	191 (49.7%)	56 (51.9%)	<0.001

Characteristic	Overall (N=1,186)	P1 (n=694)	P2 (n=384)	P3 (n=108)	p-value
Black race, n (%)	89 (7.5%)	50 (7.2%)	29 (7.6%)	10 (9.3%)	
Other race, n (%)	173 (14.6%)	95 (13.7%)	56 (14.6%)	22 (20.4%)	
Unknown race, n (%)	228 (19.2%)	100 (14.4%)	108 (28.1%)	20 (18.5%)	
Admission details					
Emergency admission, n (%)	830 (70.0%)	469 (67.6%)	307 (79.9%)	54 (50.0%)	<0.001
Observation admission, n (%)	190 (16.0%)	140 (20.2%)	35 (9.1%)	15 (13.9%)	
Urgent admission, n (%)	121 (10.2%)	66 (9.5%)	26 (6.8%)	29 (26.9%)	
Surgical same-day admission, n (%)	32 (2.7%)	15 (2.2%)	12 (3.1%)	5 (4.6%)	
Direct emergency / other, n (%)	13 (1.1%)	4 (0.6%)	4 (1.0%)	5 (4.6%)	
Etiology and evidence					
NSAH evidence level 1, n (%)	423 (35.7%)	252 (36.3%)	99 (25.8%)	72 (66.7%)	<0.001
NSAH evidence level 2, n (%)	529 (44.6%)	305 (43.9%)	190 (49.5%)	34 (31.5%)	
NSAH evidence level 3, n (%)	234 (19.7%)	137 (19.7%)	95 (24.7%)	2 (1.9%)	
Aneurysm diagnosis, n (%)	87 (7.3%)	53 (7.6%)	31 (8.1%)	3 (2.8%)	0.157
Aneurysm-securing procedure, n (%)	234 (19.7%)	137 (19.7%)	95 (24.7%)	2 (1.9%)	<0.001
ICU interventions (0–48 h)					
Nimodipine, n (%)	675 (56.9%)	432 (62.2%)	226 (58.9%)	17 (15.7%)	<0.001
EVD or ICP monitoring, n (%)	322 (27.2%)	119 (17.1%)	188 (49.0%)	15 (13.9%)	<0.001
Vasopressor use, n (%)	252 (21.2%)	43 (6.2%)	154 (40.1%)	55 (50.9%)	<0.001
Mechanical ventilation, n (%)	431 (36.3%)	120 (17.3%)	249 (64.8%)	62 (57.4%)	<0.001

Characteristic	Overall (N=1,186)	P1 (n=694)	P2 (n=384)	P3 (n=108)	p-value
Fluid balance, median [IQR], L	1.38 [-0.45–2.96]	0.94 [-0.65–2.26]	2.26 [0.22–3.79]	2.41 [-0.04–6.19]	<0.001
CRRT, n (%)	10 (0.8%)	0 (0.0%)	1 (0.3%)	9 (8.3%)	<0.001
Outcomes					
In-hospital mortality, n (%)	235 (19.81%)	44 (6.34%)	125 (32.55%)	66 (61.11%)	<0.001
ICU mortality, n (%)	182 (15.35%)	25 (3.60%)	102 (26.56%)	55 (50.93%)	<0.001
Early anemia (0–48 h), n (%)	315 (26.56%)	84 (12.10%)	159 (41.41%)	72 (66.67%)	<0.001
RBC transfusion (0–48 h), n (%)	24 (2.02%)	3 (0.43%)	13 (3.39%)	8 (7.41%)	<0.001
ICU LOS, median [IQR], days	6.96 [2.88–13.41]	6.38 [2.71–10.50]	10.56 [3.83–17.50]	5.38 [2.10–12.19]	<0.001
Hospital LOS, median [IQR], days	11.60 [6.42–19.91]	9.94 [6.51–15.24]	15.85 [6.70–24.79]	13.31 [3.46–28.29]	<0.001

ESM Table 8. Early physiological profiles of identified phenotypes

Physiological variable (0–48 h)	Overall (N=1,186)	P1 (n=694)	P2 (n=384)	P3 (n=108)	p-value
Hemoglobin min, g/dL	11.10 [9.80–12.40]	11.70 [10.80–12.80]	10.25 [9.10–11.50]	8.85 [7.38–10.43]	<0.001
GCS motor min	5.00 [1.00–6.00]	6.00 [5.00–6.00]	1.00 [1.00–4.00]	1.00 [1.00–5.00]	<0.001
MAP min, mmHg	61.00 [54.00–67.00]	64.00 [58.00–70.00]	58.00 [51.00–63.00]	55.00 [47.50–61.00]	<0.001
Shock index max	0.88 [0.77–1.05]	0.82 [0.73–0.92]	1.00 [0.87–1.17]	1.16 [0.93–1.42]	<0.001
SpO2 min, %	92.00 [90.00–94.00]	92.00 [90.00–94.00]	93.50 [92.00–96.00]	88.50 [78.00–92.00]	<0.001
Creatinine max, mg/dL	0.80 [0.70–1.10]	0.80 [0.60–0.90]	0.90 [0.70–1.10]	1.90 [1.18–3.20]	<0.001
INR max	1.20 [1.10–1.30]	1.10 [1.10–1.20]	1.20 [1.10–1.30]	1.80 [1.40–2.30]	<0.001
Platelets min, $\times 10^3/\mu L$	188.0 [150.0–232.0]	204.0 [169.25–245.0]	174.0 [140.0–213.5]	92.0 [40.75–160.25]	<0.001

ESM Table 9. Multivariable logistic regression models for in-hospital mortality

Variable	Model 1 aOR (95% CI)	Model 1 p-value	Model 2 aOR (95% CI)	Model 2 p-value
Phenotype 2 vs P1	7.59 (5.07–11.36)	<0.001	4.02 (2.60–6.24)	<0.001
Phenotype 3 vs P1	21.21 (12.08–37.26)	<0.001	11.75 (6.52–21.20)	<0.001
Early anemia	0.99 (0.68–1.44)	0.955	1.01 (0.69–1.50)	0.941
Age, per year	1.03 (1.02–1.04)	<0.001	1.03 (1.02–1.04)	<0.001
Male sex	1.21 (0.86–1.70)	0.274	1.15 (0.81–1.63)	0.430
Aneurysm diagnosis	0.52 (0.24–1.14)	0.101	0.55 (0.25–1.24)	0.148
NSAH evidence level 2	0.88 (0.60–1.29)	0.499	0.91 (0.59–1.42)	0.686
NSAH evidence level 3	0.65 (0.38–1.09)	0.102	0.78 (0.000–inf)	1.000
Emergency admission	2.13 (0.75–6.06)	0.156	2.53 (0.76–8.38)	0.129
Observation admission	1.81 (0.59–5.54)	0.298	2.21 (0.58–8.43)	0.244
Urgent admission	3.38 (1.11–10.33)	0.033	3.63 (0.95–13.95)	0.060
Nimodipine	—	—	0.52 (0.46–0.58)	<0.001
Vasopressor use	—	—	2.63 (1.74–3.98)	<0.001
Mechanical ventilation	—	—	2.09 (1.40–3.11)	<0.001
RBC transfusion	—	—	0.39 (0.13–1.17)	0.094
CRRT	—	—	0.80 (0.17–3.72)	0.781
EVD or ICP monitoring	—	—	1.41 (1.00–1.99)	0.052
Fluid balance, L	—	—	0.97 (0.92–1.02)	0.233

Model 1 adjusted for baseline demographics, admission type, NSAH evidence level, aneurysm diagnosis, and early anemia. Model 2 additionally included process-of-care variables and is exploratory.

ESM Table 10. Cox proportional hazards models for in-hospital survival

Variable	Unadjusted HR (95% CI)	p-value	Clinical-adjusted HR (95% CI)	p-value	Process-adjusted HR (95% CI)	p-value
Phenotype 2 vs P1	4.20 (2.97–5.94)	<0.001	4.45 (3.11–6.37)	<0.001	3.08 (2.09–4.54)	<0.001

Variable	Unadjusted HR (95% CI)	p-value	Clinical-adjusted HR (95% CI)	p-value	Process-adjusted HR (95% CI)	p-value
Phenotype 3 vs P1	7.94 (5.38–11.70)	<0.001	8.62 (5.50–13.52)	<0.001	5.24 (3.25–8.47)	<0.001
Early anemia	—	—	0.76 (0.56–1.01)	0.062	0.73 (0.54–0.99)	0.044
Age, per year	—	—	1.02 (1.01–1.03)	<0.001	1.02 (1.01–1.03)	<0.001
Male sex	—	—	0.96 (0.74–1.26)	0.788	0.88 (0.67–1.16)	0.365
Nimodipine	—	—	—	—	0.59 (0.40–0.87)	0.007
Vasopressor use	—	—	—	—	2.15 (1.57–2.94)	<0.001
Mechanical ventilation	—	—	—	—	1.79 (1.29–2.50)	<0.001

The process-adjusted Cox model includes 0–48 h interventions as fixed exploratory contextual covariates. It is vulnerable to treatment-selection and immortal time bias and should not be interpreted as estimating treatment effects.

ESM Table 11. eICU external validation cohort characteristics and outcomes

Characteristic	Overall (N=843)	P1 (n=539)	P2 (n=222)	P3 (n=82)
Age, median, years	60.0	59.0	61.0	63.0
Hemoglobin min, g/dL	11.4	11.9	10.7	9.5
GCS motor min	6.0	6.0	4.0	4.0
MAP min, mmHg	58.0	62.0	51.0	51.5
Shock index max	0.97	0.88	1.14	1.26
SpO2 min, %	90.0	90.0	92.0	74.5
Creatinine max, mg/dL	0.81	0.75	0.81	1.54
INR max	1.1	1.1	1.1	1.5
Platelets min, $\times 10^3/\mu L$	200.0	207.0	202.5	146.0
APACHE score, median	43.0	36.0	57.0	79.0
APS score, median	33.0	27.0	49.0	67.0
	0.112	0.069	0.243	0.426

Characteristic	Overall (N=843)	P1 (n=539)	P2 (n=222)	P3 (n=82)
Predicted hospital mortality, median				
Early anemia (0–48 h), n (%)	183 (21.7%)	60 (11.4%)	76 (34.5%)	47 (58.0%)
RBC transfusion (0–48 h), n (%)	24 (2.8%)	2 (0.4%)	13 (5.9%)	9 (11.0%)
Hospital mortality, n (%)	121 (14.35%)	29 (5.38%)	57 (25.68%)	35 (42.68%)
ICU mortality, n (%)	65 (7.71%)	15 (2.78%)	36 (16.22%)	24 (29.27%)

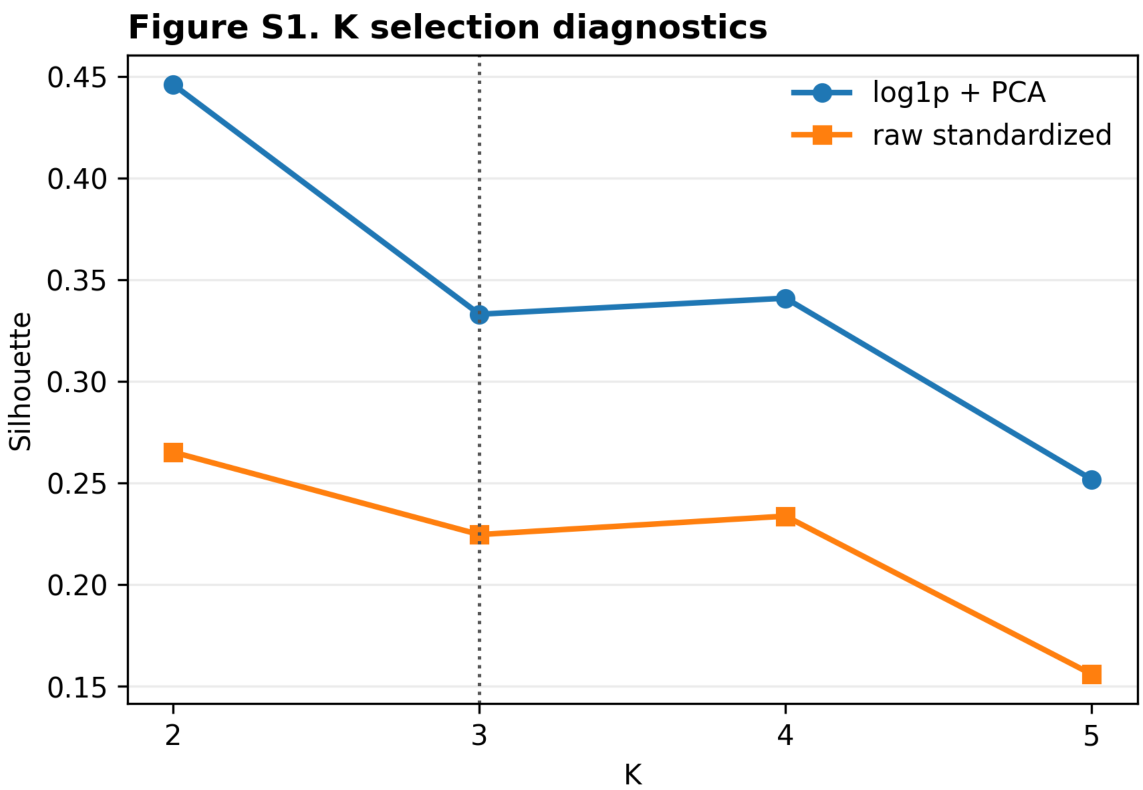
ESM Table 12. Sensitivity analysis summary

Analysis	N	P1 mortality	P2 mortality	P3/P4 mortality	Minimum subgroup N
MIMIC primary log-PCA	1,186	6.34%	32.55%	61.11% (P3)	108
eICU frozen transport	843	5.38%	25.68%	42.68% (P3)	82
MIMIC complete case	1,120	6.53%	38.58%	65.07% (P3)	63
MIMIC strict aneurysm	763	5.73%	30.82%	50.00% (P3)	48
MIMIC ICU LOS ≥48 h	1,005	6.07%	34.38%	58.70% (P3)	46
MIMIC no RBC 0–48 h	1,162	6.58%	37.47%	68.85% (P3)	61
MIMIC Hb-free clustering	1,186	6.85%	39.52%	64.10% (P3)	78
MIMIC INR-free clustering	1,186	6.00%	35.32%	66.67% (P3)	84
MIMIC K=4 resolution	1,186	6.17% (P1)	37.99% (P2)	61.54% (P3) / 56.41% (P4)	39
eICU ICU LOS ≥48 h	626	6.9%	24.2%	33.9% (P3)	62
eICU no RBC 0–48 h	819	5.4%	25.8%	39.7% (P3)	73
eICU strict SAH	605	6.6%	30.5%	45.6% (P3)	68
eICU INR-free transport	868	4.6%	24.4%	38.7% (P3)	124

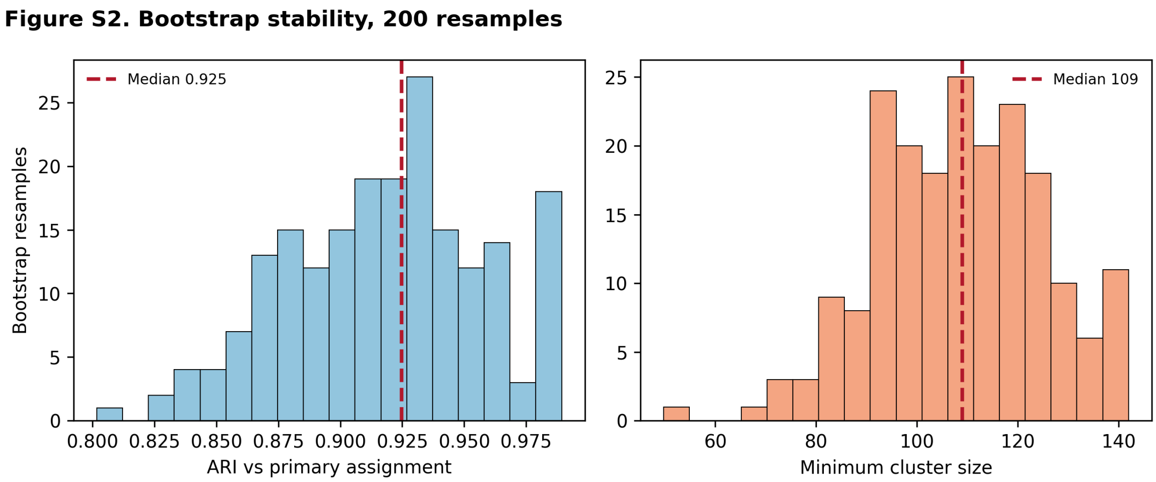
ESM Table 13. Bootstrap stability analysis

Metric	Mean	Standard deviation	Minimum	Maximum
Adjusted Rand Index (ARI)	0.9199	0.0479	0.7454	0.9927
Same ordered label rate	96.96%	1.97%	89.21%	99.75%
Minimum cluster size, N	108.7	16.8	48.0	140.0

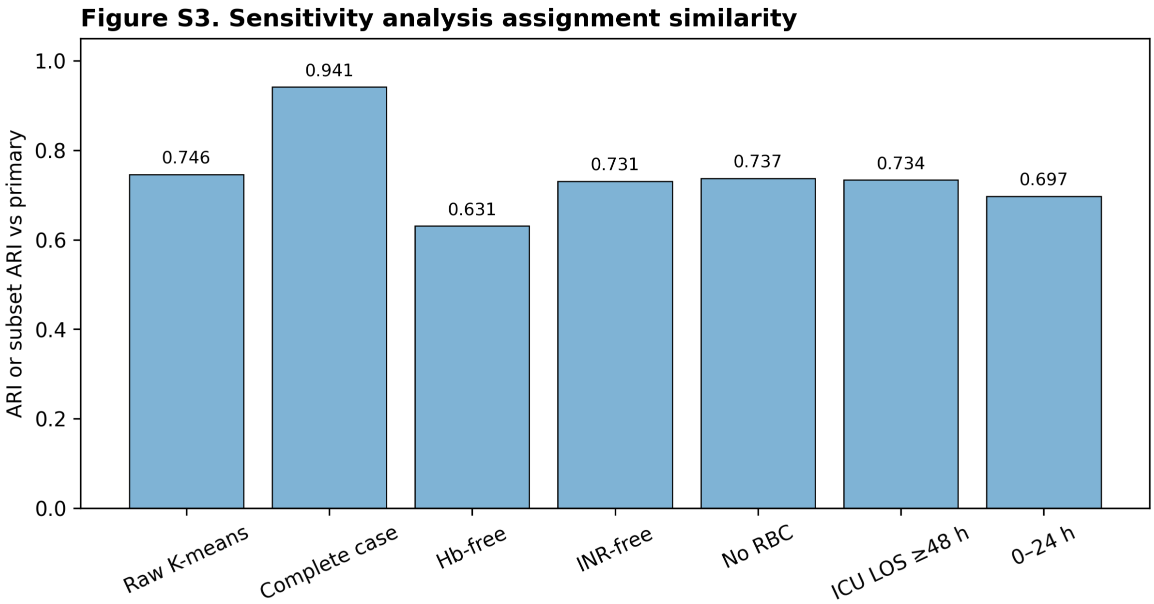
ESM 4. Supplementary Figures



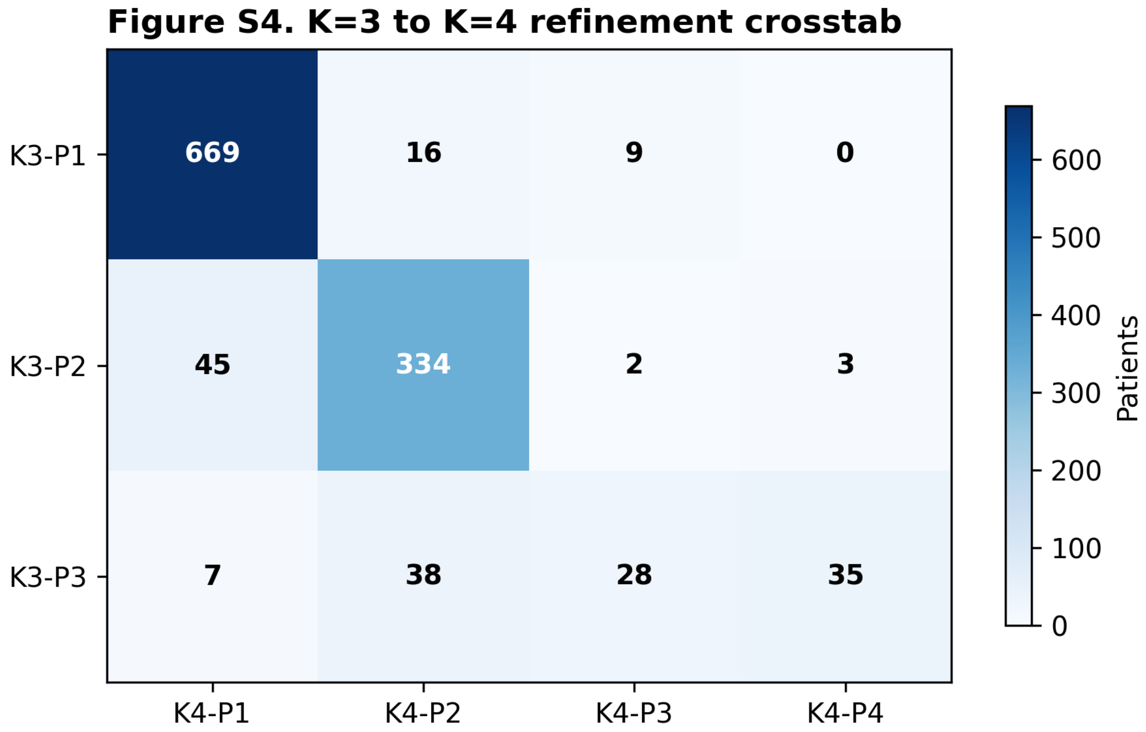
ESM Fig. 1. Cluster-quality metrics across K=2 to K=5 for raw standardized and primary log-PCA pipelines.



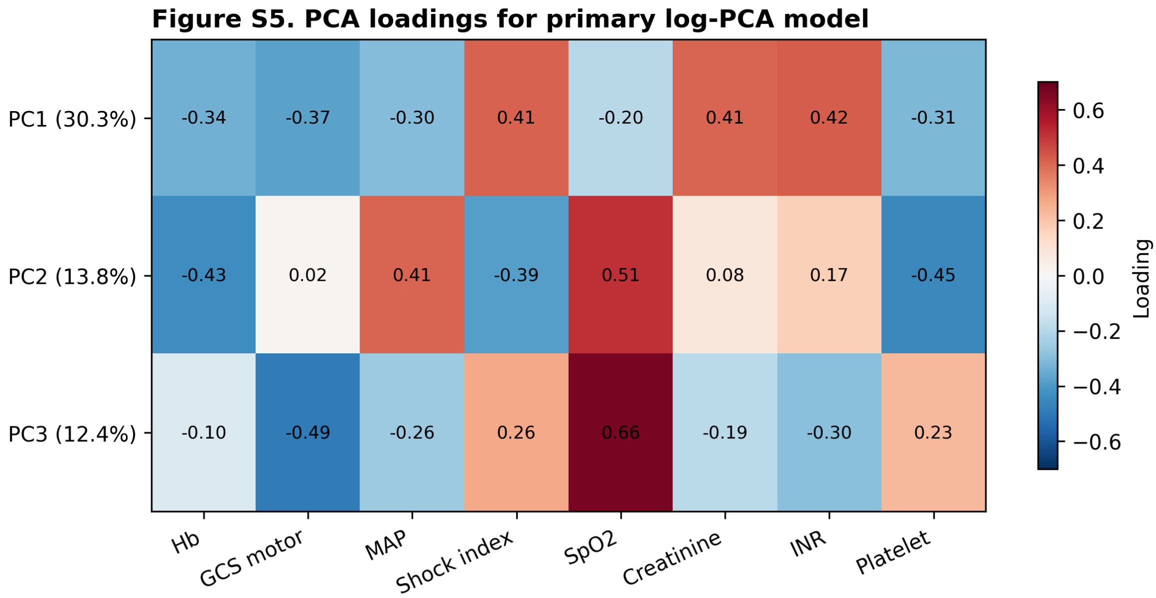
ESM Fig. 2. Adjusted Rand Index distribution across 200 bootstrap iterations compared with the primary K-means K=3 partition.



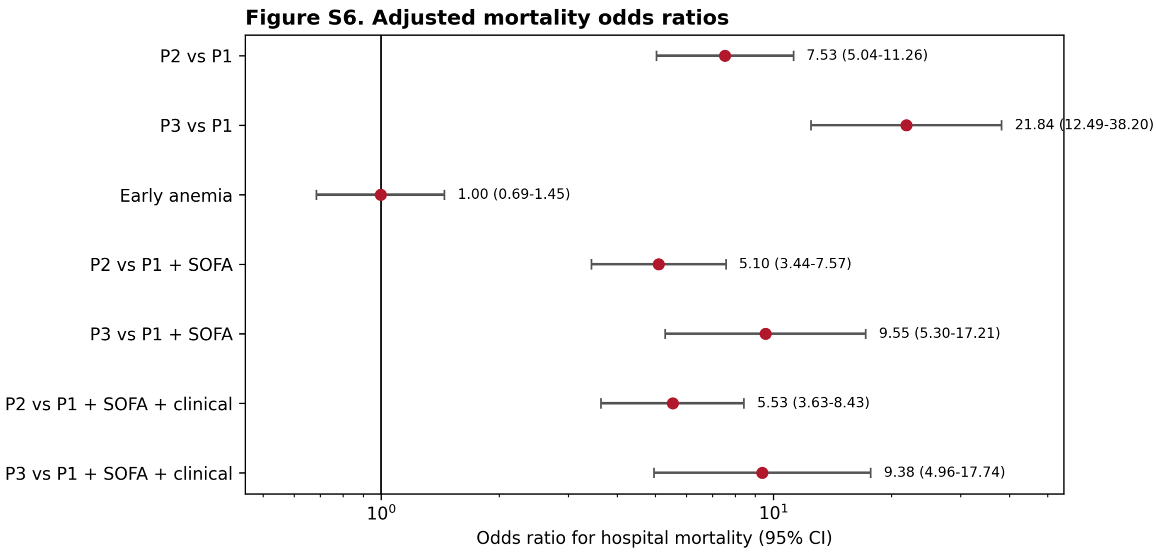
ESM Fig. 3. Standardized cluster centers for sensitivity partitions, demonstrating alignment with the primary log-PCA centroids.



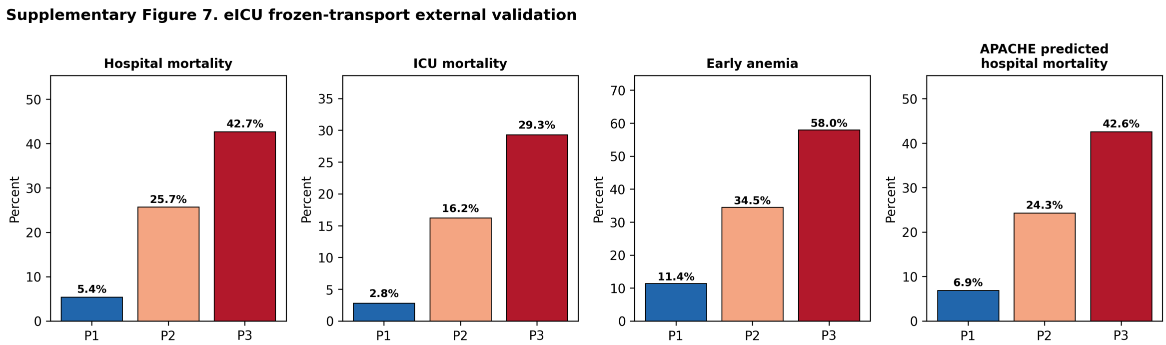
ESM Fig. 4. High-resolution K=4 refinement crosstab, illustrating how the primary P3 phenotype splits into two smaller high-risk subcohorts.



ESM Fig. 5. Loadings of the eight physiological variables on the first three principal components.



ESM Fig. 6. Forest plot showing multivariable logistic regression and Cox proportional hazards odds/hazard ratios with 95% confidence intervals for primary and sensitivity models.



ESM Fig. 7. eICU validation performance, including transported phenotype calibration, calibration slope, and de novo clustering comparison metrics.

ESM 5. Reproducibility Notes

Analysis scripts

Step	Script or artifact	Purpose
Cohort SQL execution	`scripts/12_run_non_traumatic_sah_cohort_sql.py`	Runs `10_create_non_traumatic_sah_cohort.sql` in BigQuery
Phenotype analysis	`scripts/13_run_non_traumatic_sah_analysis.py`	Executes `11_bigquery_notebook_non_traumatic_sah_analysis.py` locally

Step	Script or artifact	Purpose
Figure generation	`scripts/generate_manuscript_figures.py`	Regenerates all main and supplementary figures
PDF rendering	`scripts/convert_manuscript_to_pdf.py`	Converts manuscript and ESM markdown files to PDF
Cohort-flow counts	`dist/20260708/figures/fig1_cohort_flowchart_data.json`	Stores BigQuery-derived MIMIC and eICU flow counts

Key reproducibility parameters

Component	Value
Primary physiology window	0–48 h after ICU admission
Primary feature count	8 variables
Missing-data rule	≤2 missing core features
Imputation	MIMIC-IV derivation-cohort median; frozen for eICU
Skewed-feature transform	log1p for creatinine and INR
Standardization	MIMIC-IV derivation-cohort mean and standard deviation
PCA components retained	3
Variance explained	56.41%
K-means K	3
K-means random state	42
K-means n_init	100
Bootstrap iterations	200
External validation	Frozen MIMIC medians, scaler, PCA loadings, and centroids applied to eICU

BigQuery provenance

The project-level BigQuery dataset for the current non-traumatic SAH study is `mimic-study-498508.non_traumatic_sah_study`. MIMIC-IV source tables were drawn from PhysioNet MIMIC-IV 3.1 datasets. eICU validation used the eICU-CRD source tables available through PhysioNet. The exact final table names should be verified after the final BigQuery rerun and recorded in the repository release notes before journal submission.